Cazuri → schimbăm un procesor cu unul mai rapid
→ adăugăm mai multe procesoare

① $CPU_{time}$ = Nr. of clock cycles per program × clock cycle time

= Instructions count × Clock cycles per instruction × clock cycle time

Prescurtări = IC × CPI × CCT

② $CPI = \sum \frac{IC_i}{IC} \times CPI_i$ ⟶ $IC_i$ - nr. of instr. in class i

$CPI_i$ – CPI of class i

③ Performance of X $= \frac{1}{\text{Execution time } X}$

ex: A is faster n times than B

$\Rightarrow n = \frac{\text{Performance } A}{\text{Performance } B} = \frac{\text{Execution time } B}{\text{Execution time } A}$

## Other metrics

MIPS – million instructions per second

① $MIPS = \frac{\text{Instructions Count}}{\text{Execution Time} \cdot 10^6} = \frac{\cancel{IC}}{\cancel{IC} \cdot CPI \cdot CCT \cdot 10^6} = \frac{CR}{CPI \cdot 10^6}$

↳ $CPU_{time}$

② $10^0$ $10^3$ kilo $10^6$ mega $10^9$ giga $10^{12}$ tera $10^{15}$ peta $10^{18}$ exa $10^{21}$ zetta $10^{24}$ yotta

$10^{-24}$ yotto $10^{-21}$ zepto $10^{-18}$ atto $10^{-15}$ femto $10^{-12}$ pico $10^{-9}$ nano $10^{-6}$ micro $10^{-3}$ mili $10^0$

③ $GFLOPS = \frac{IC_{FP}}{CPU_{time} \cdot 10^9}$

④ Relative $MIPS_X = \frac{\text{Execution time}_{\text{Reference}}}{\text{Execution time } X} \cdot \text{MIPS Reference}$

⑤ Normalised $GFLOPS = \frac{\text{Normalized } IC_{FP}}{CPU_{time} \cdot 10^9}$

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⑥ $speedup_{ov} = \frac{1}{(1 - \text{fr. opt}) + \frac{\text{fr opt}}{\text{speedup}}}$

1. A: CCT = 250 ps ; CPi = 2

B: CCT = 500 ps ; CPi = 1,2

same iC

Which is faster

$CPU_{time\,A} = iC \cdot 500\,ps$

$CPU_{time\,B} = iC \cdot 600\,ps$

$\frac{CPU_{time\,B}}{CPU_{time\,A}} = 1,2 \Rightarrow$ A is 1,2 faster than B

2.

| | A | B | C |
|---|---|---|---|
| CPi | 1 | 2 | 3 |

| iC | A | B | C | |
|---|---|---|---|---|
| 1 | 2 | 1 | 2 | = 5 |
| 2 | 4 | 1 | 1 | = 6 |

2 seg. de cod

$CPU_{time\,1} = 5 \cdot \left( \frac{\overset{iC_i\ CPI_i}{2 \cdot 1 + 1 \cdot 2 + 2 \cdot 3}}{5} \right) \cdot CCT = 10\ CC \cdot CCT$

$CPU_{time\,2} = 6 \cdot \frac{4 \cdot 1 + 1 \cdot 2 + 1 \cdot 3}{6} \cdot CCT = 9\,cc \cdot CCT$

$\frac{Perf\,2}{Perf\,1} = \frac{CPU_{time\,1}}{CPU_{time\,2}} = \frac{10}{9} \simeq 1,1$

3. compilator → vechi
→ nou

$iC_{new} = 0,6 \cdot iC_{old}$

$CPI_{new} = 1,1 \cdot CPI_{old}$

same CCT → acelaşi computer

$CPU_{time\,old} = 15\ sec$

$CPU_{time\,new} = \overbrace{0,6 \cdot iC_{old} \cdot 1,1 \cdot CPI_{old} \cdot CCT}^{CPU_{time\,old}} = 15 \cdot 0,6 \cdot 1,1 = 9,9\ sec$

4. · frequency of FP op = 25%
· average CPI of FP op = 4
· average CPI of others op = 1,33
· frequency of FPSQR = 2%
· average CPI of FPSQR = 20

Which option is the best
- decrease CPI of FPSQR to 2
- decrease the avg CPI of FP to 2,5

$CPU_{time_1}$ vs $CPU_{time_2}$ $\Leftrightarrow$ $CPI_1$ vs $CPI_2$

$$CPI_{original} = \underbrace{0,25}_{freq} \cdot \underbrace{4}_{average} + \underbrace{0,75}_{restul} \cdot 1,33 \approx 2\ cc$$

$$CPI_1 = CPI_{original} - \underbrace{\frac{2}{100}}_{freq\ FPQSR}(20 - 2) = 2 - 0,02 \cdot 18 \approx 1,64\ cc$$

$$CPI_2 = 0,25 \cdot 2,5 + 0,75 \cdot 1,33 \approx 1,625\ cc.$$

$\Rightarrow$ second option is better

5.

| $CPU_A$ | $CPU_B$ |
|---|---|
| CMP $r_1, r_2$ | BEQ $r_1, r_2$, FOO |
| BEQ FOO | ⋮ |
| ⋮ | FOO |
| FOO: ADD $r_6, r_5, r_6$ | |
| ⋮ | |

A: $CPI_{Br} = 2\ cc \rightarrow CPI_{oth} = 1$

20%

$CCT_B = 1,25 \cdot CCT_A$

$IC_B = 0,8 \cdot IC_A$

| A: | B: |
|---|---|
| 20% B | 25% B |
| 20% C | |
| 60% O | |

$$CPU_{time_A} = CCT_A \cdot IC_A \cdot CPI_A$$

$$CPI_A = \underbrace{20\%}_{branch\ conditional} \cdot 2 + 80\% \cdot 1 \approx 1,2\ cc$$

$\Rightarrow CPU_{time_A} = 1,2 \cdot IC_A \cdot CCT_A$

$$CPU_{time_B} = IC_B \cdot CPI_B \cdot CCT_B = 0,8 \cdot IC_A \cdot CPI_B \cdot 1,2 \cdot CCT_A$$

$$CPI_B = 0,25 \cdot 2 + 0,75 \cdot 1 = 1,25\ cc$$

$\Rightarrow 1,25 \cdot IC_A \cdot CCT_A$

$\Rightarrow$ nu e un plus optimizarea B

6.

| INSTR | Freq | CPI |
|---|---|---|
| ALU | 43% | 1 |
| Load | 21% | 2 |
| Store | 12% | 2 |
| Branch | 24% | 2/3 |

L/S → 25% ALU with one operand from the memory

LBR $h_1$, $[h_2]$ ABS $h_3$, $[h_2]$, $h_4$

ABS $h_3$, $h_1$, $h_4$ new ALU → CPI = 2cc CPI new branch = 3 c.c.

$CPU_{time_{L/S}} = IC_{L/S} \cdot CPI_{L/S} \cdot CCT_{L/S} = IC_{L/S} \cdot 1{,}57 \cdot CCT_{L/S}$ neoptimizat

$CPI_{L/S} = 0{,}43 \cdot 1 + \underbrace{0{,}57}_{\text{others } (0{,}24+0{,}21+0{,}12)} \cdot 2 = 1{,}57$ c.c.

$CPU_{time_{new}} = IC_{new} \cdot CPI_{new} \cdot CCT_{new}$ L/S

$IC_{new} = IC_{L/S} \cdot (1 - 0{,}25 \cdot 0{,}43)$

$$CPI_{new} = \frac{\overset{\text{w/o mem.}}{0{,}43 \cdot (1-0{,}25) \cdot 1} + \overset{\text{w mem.}}{0{,}43 \cdot 0{,}25 \cdot 2} + \overbrace{(0{,}21 - 0{,}43 \cdot 0{,}25)}^{\text{Load}} \cdot 2 + \overset{\text{Store}}{0{,}12} \cdot 2 + \overset{\text{Branch}}{0{,}24} \cdot 3}{(1 - 0{,}25 \cdot 0{,}43)} =$$

$\approx 1{,}7$

$CPU_{time_{new}} = IC_{L/S} \cdot CCT_{L/S} \cdot 1{,}7$

7. Optimized 50% of ALU

$CCT = 20ms$ + date 6

$CR = \frac{1}{CCT} = 50$ MHz $CPI_{orig} = 0,43 \cdot 1 + 0,57 \cdot 2 = 1,57$

$CPU_{time\ L/S} = IC_{L/S} \cdot 1,57 \cdot 20 = IC_{L/S} \cdot 31,4$ ms

$MIPS_{L/S} = \frac{50 \cdot 10^6}{1,57 \cdot 10^6} = 31,85$

$CPU_{time\ opt} = IC_{co} \cdot CPI_{co} \cdot 20$

$CPI_{opt.} = \frac{\left(0,43 \cdot \frac{50}{100}\right) \cdot 1 + 0,57 \cdot 2}{1 - 0,43 \cdot 0,5} = 1,73$

$\Rightarrow CPU_{opt} = IC_{L/S} \cdot (1 - 0,43 \cdot 0,5) \cdot CPI_{opt} \cdot 20 = IC_{L/S} \cdot 27,1$ ms

$MIPS = \frac{50}{CPI_{opt}} = 28,9$

Uneori MIPS variază i.p. cu $CPU_{time}$